

Big Data Analytics and Visualization of TV Shows Using TVmaze API

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Activity 1

Introduction to Big Data

Big data represents a large amount of rows and columns that are continuously updated, and every part or feature of the data has a large number of examples; this type of data has specific ways to represent, manage, handle, and analyze it can not get treated like traditional data. Developing new technologies and then using them in many organizations, like platforms, sensors, social media, and business systems, result in collecting large-scale data that is rapidly updated. Big data is not just large stored data its also defined by the speed at which data is produced and how valuable and reliable in analysis and for generating decisions. Organisations adopt specific, effective modern technologies to turn the raw quantitative or qualitative data into valuable insights for the organisation's services, updates, and activities.

Characteristics of Big Data

The characteristics of Big Data are commonly explained through the five Vs: Volume, Velocity, Variety, Veracity, and Value. These features are connected to the importance of the big dataset; every feature should be involved to raise the value of the insights, enhance the analytics phase, and make it easy to choose the needed technologies to handle

Volume is defined as the large number of rows and columns filled with information, representing an enormous volume of stored data. The main source of this large amount of data is often transactions, websites, mobile applications, customer interactions, sensors, social media, and other digital systems; manually stored data would never be as large as the output from these techniques. The collected data from these continuously generating techniques creates challenges related to data storage, processing, and management. This requires organisations to continuously clean and use the data to analyse and provide scalable technologies that can handle large datasets efficiently.

Velocity is the feature that distinguishes big data from other types of data because of its speed of generation and collection from its sources. Organisations with systems or platforms to provide services may need to analyze the generated data continuously to make timely decisions. The modern data-generating technologies need effective methodologies and platforms to store and process the information in real time or near real time, which would enhance the organization's activities and services by providing the right decisions based on the latest organizational updates.

Variety refers to generating and collecting different types of data because traditional data is always on the same scale and type, but big data contains semi-structured data and structured data at the same time because its live generating data from all the organization's services and activities. So the same organization's large data may contain text, images, videos, audio files, social media posts, and documents. Managing this variety allows organisations to combine information from different sources and gain a more complete understanding of their activities and customers.

Veracity in big data is not just a large amount of stored data; the data has to be meaningful and have accuracy, reliability, consistency, and quality. Large datasets can contain missing values, duplicated records, non-scaled values, incorrect information, or inconsistencies. So veracity is a main feature in big

data for organizations. When organizations are in the decision-making process and they are relying on poor-quality data, it would be hard to analyze and produce insights; it may be wrong, which will affect the quality of the decision. Data cleaning, validation, and quality-control processes are therefore important parts of Big Data management.

Value is defined by how useful the data is to produce meaningful and high-quality decisions and how easy to analyze and find patterns or trends because of connected relationships between the dataset values. Because large quantities of information are not enough to categorise the dataset as a useful dataset. After obtaining a valuable dataset and identifying patterns, trends, and relationships, this will enhance the organization's services and activities and improve its processes

Value of Big Data for Organisations

Using the organisation's big data provides direct communication with company updates by collecting and analyzing the information that comes from the big data, because big data will enable the organisation to identify trends, patterns, and relationships. Figuring out the insights and modelling them to the management department will enhance the organisation's activities and services by reducing the costs of errors and identifying opportunities. Big data also enhances and supports the customer overview and experience because analyzing the data and the customers' behaviour will provide a better understanding of their needs and customize the service based on the insights. As a result, organisations can make evidence-based and insights ideas decisions rather than relying mainly on assumptions and manual thinking or personal judgement, which can improve performance and competitiveness.

Big Data and Decision-Making

Big Data takes the decision-making process to another level because decisions will be based on evidence and investigate the reason behind every activity noticed in the data, changing also decision for the future based on predictions related to specific trends or patterns. Descriptive analytics can show what has happened, diagnostic analytics can help explain why it happened, and predictive analytics uses the relationships in the dataset to predict what will happen in the future and enhance resource allocation by taking a decision based on real-life evidence. By using all the analytics steps to understand what's behind the outcome and make predictions based on it, organisations can evaluate different situations and make more informed strategic, well-aligned, and operational decisions that will enhance the organisation's services and improve its development. However, the effectiveness of these decisions depends on the quality, accuracy, and relevance of the data, because poor-quality data can result in misleading insights and inappropriate decisions.

Value for End Users

Using big data and understanding the dataset plays a big role in the end user's experience because analysing users' behaviour and understanding their preferences will allow and light the organization to provide more personalized and customized service or product. Such as, organizations that provide their service through a digital platform might analyze users behaviour during real time using the platform and suggest products, services, and content based on their preferences. Big Data can also improve the speed and quality of services. Also, the continuous analyzing for the system's big data will allow companies to detect problems and errors what will prevent these problems that users face. This can lead to shorter waiting times, improved service quality, and greater customer satisfaction.

Statistical Techniques

Descriptive analytics is a statistical approach employed in describing and summarising existing information in order to comprehend what has already occurred. This approach entails the use of

metrics such as the average, median, percentage, frequency, and totals in order to highlight patterns within the data set. For instance, an organization that is working with TV show information may compute the average ratings, average duration, and the number of shows based on genres.

Mean and Standard Deviation: Mean and standard deviation are the best techniques to understand and get an overview of large-scale numerical data. The mean provides the average value of a category or a pattern, while standard deviation measures how much the values change around the average; that is what creates a relationship between those two statistical measurements. Applying these two measurements in large-scale data summarizes the dataset information, provides the centre and spread of data, and supports decision-making because they analyze the big data effectively. For example, an organisation could calculate the average value and variation of a business measure across millions of records to identify general patterns and differences and specify the changes over time based on the average.

Correlation Analysis: Using correlation analysis to understand the relationship between values in large-scale data to understand the dataset direction and dependencies between columns. It measures the strength of a linear relationship; the results indicate the type of relationship between values, so a positive relationship indicates that the two values increase together, and a negative result indicates that one value increases and the other decreases. So correlation analysis can help organizations understand the data and the relationships without huge effort due to its scale what will benefit predictive modeling in selecting useful features in the data

Graphical and Visualisation Techniques

Bar Charts: A bar chart is a visual format used to simplify understanding the comparison of values between two categories, which makes it quick to analyze and access instead of dealing with thousands or millions of values and records. To simplify the graph pattern, the data should be aggregated into suitable categories and discover relationships then represent it using bars. For example, an organisation can use a bar chart to compare the number of customers, transactions, or products across different categories or regions.

Line Charts: A line chart is a graphical format to represent the changing of values in a large-scale data set during a specific period of time; it can be aggregated into years, months, or days. The line is a connected set of data points over time, which allows the viewer to understand the changes between increases and decreases, fluctuations and trends. Organisations can use line charts to monitor measures such as sales, website traffic, demand, or operational performance and use these trends to support forecasting, planning, and decision-making.

Interactive Dashboards: An interactive dashboard is a visualisation technique that contains a header that is related to the use case and multiple charts and figures with different conditions and parameters about the same use case or a problem to simplify the decision making process. Unlike static visualisations, users can interact with the dashboard by applying filters, selecting categories, or changing time periods to explore specific aspects of a dataset, which lets them understand the data more from every field. For example, an organisation analysing TV show data can use an interactive dashboard to filter shows by genre, year, rating, or country and examine patterns in ratings, runtime, and popularity. This simplify to the decision makers to understand the most important features that affect the rating of the shows

Software Tools

Polars: Polars perform an efficient tool for handling large-scale data, and it also provides the best environment for data manipulation and analysis because it uses parallel processing, an optimised query engine, and supports both eager and lazy evaluation. It doesn't overload the system by uploading the whole dataset at once. Compared with pandas, which has memory limitations for loading medium to large datasets and may cause the whole process to crash, which makes Polars is more efficient for large datasets to analyze and produces results without taking long time

Apache Spark (PySpark): Apache Spark is a tool that works across multiple machines based on the problem, and there handle the large datasets and analyze it using a specific machine or across machines. PySpark is a framework that analyzes data with Python in a parallel cluster. This is used when the dataset is too large to manage it on a single machine. PySpark can be used for data manipulation, large-scale analytics, and machine learning, making it an important technology for organisations working with distributed Big Data environments.

Roles of Data Specialists

Data Specialist Role	Main Focus	Main Responsibilities	Key Challenges
Data Engineer	Data collection, processing, and infrastructure	Responsible for data collection, cleaning missing values, inconsistencies, and duplicates, and then transforming the data. Develops data pipelines and ETL processes, and deploys Big Data technologies so analysts and data scientists can use the data.	Handling large-scale data, ensuring reliability of the data for each record, controlling system cost performance, and maintaining the data pipelines which will affect the organization performance
Data Scientist	Advanced analytics and machine learning	This role is responsible for applying statistical techniques to take an overview of the data to determine the problems and how it needs to be handled; also applying machine learning techniques to generate insights to provide storytelling and reporting.	The data used for the applied techniques should be high quality and cleaned to ensure accurate results, also tuning model performance to achieve the best results
Business Analyst	Business problems and decision-making	Identifies and analyses the business problems based on the stored data, reports technical and non-technical problems, generate analysis to identify the business risks and benefits	Identifies the business requirements based on the complex data, reporting the technical updates to non-technical team members and identifying the business impact of solutions.

Data Compliance Strategies

Strategy	How it Maintains Compliance	Effectiveness	Gaps / Practical Limitations	Actionable Refinement
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Data Protection, Informed Consent and Privacy	The data holder should protect the users' data and inform them how their data is used and collected	Applying this strategy as a data specialist is highly effective in compliance with the regulations because it provides transparency for the users. What reduces its effectiveness is that it depends on the users' understanding of the provided information	The risk of this strategy begins when the data is collected from many sources and used for different purposes, so it's difficult to update users on how it was collected or used	Provide clear information for all user types, age groups, qualifications, and regularly apply consent procedures
Compliance Management Software	Using a software tool that complies with all the privacy regulations, doesn't perform any illegal activity, and manages and monitors the compliance requirements	It's highly effective because its provide automation of compliance and reject the non compliance without taking much effort and time for big data. It doesn't work with software that is poorly configured or with outdated existing systems, so it might not collect the needed monitoring data	It needs high implementation costs, so it might not work with startups, and it needs a continuous maintenance team to comply with all regulations	Select suitable software match with all the financial capabilities, and regularly review its compliance functions.
Identifying and Escalating Quality Risks in Data Analysis	Understand the entered data, identify data quality problems, and alert for the identified risk to take appropriate action	Its effectiveness centers on handling future problems earlier so it doesn't affect the insights and decisions. but also an error might still happen, and not all risks can be identified, so that will produce earlier problems	Large and complex datasets can contain many quality issues, making continuous identification and resolution difficult.	Use regular data-quality checks, define clear escalation procedures, and apply appropriate mitigation or corrective actions when risks are identified.

key issues data specialists encounter in a data-driven culture

Key Issue	Organisational Impact	Pros and Cons / Alternative Perspective	Evaluation & Conclusion
Data Quality and Preparation	Using poor-quality data, for example has many missing values, inconsistencies, and duplicates, has high severity on decisions because it will generate wrong insights and an unreliable analysis	Careful preparation improves big data reliability and decision quality because of having a large number of accurate examples. Preparing and processing large-scale data takes a lot of time and effort. An alternative view is using automation to complete data preparation and quality assurance	This is a high-priority issue and a very serious challenge that all large-scale data have because the impact of the decisions and how accurate they are only depends on the data behind them. Organisations should invest in automated validation and preparation where appropriate. Large Data Preparation always involves multiple formats and platforms.
Data Security and Privacy	Very high severity because if the organisation faces any data leakage or exposure, that will affect the organisation's reputation and will have	Bulding secure environment for all the organaizations data enhances performance and improves trust. However, it can create access complexity and require high costs to maintain the environment and support a regular maintenance team. An alternative perspective is that these controls are	This issue should be addressed as the first step before any data manipulation because all the stored data should be protected for the users' and organizations' benefit. Cyberattacks can leak the data and expose it if a secure environment and protection regulations aren't applied

	legal, financial, and organisational problems	necessary even if they make data management more difficult.	
Bias and Poor Data Practices	Very high severity because if the data was imbalanced and the model took biased way between data that will affect the decisions, and it won't be fair for all groups	Using multiple opinions and ways to make decisions and not relying on an analytical model for complex data increases the reliability of decisions. Bias situations are difficult to detect and identify, particularly in complex and large-scale datasets and analytical models. An alternative perspective is that there still has to be a human check on the results to identify how the models understood the data and identify rather than relying only on algorithms.	This is a real issue that should be monitored to prevent because biased analysis lead to inconsistent decisions what will affect trust. Data specialists should disclose assumptions, use appropriate algorithms during analysis, and provide human checking after results. This identifies conflicts of interest and bias from a single viewpoint as poor practices.
Multiple Data Sources	High severity because collecting and combining data from different sources is a high risk of inconsistency because of duplicates and repeated records of information. This can reduce the reliability of the final analysis.	Combining multiple sources for the same problem or case study provides wider information and more insights. Integration increases complexity and can introduce errors. An alternative perspective is that centralised platforms can make information easier to manage and access, and the data need to be cleaned and filled with appropriate values.	Combining data from many sources has many benefits in providing examples and giving wider information, but it should be properly managed by applying scaling, cleaning, and transforming techniques; it should use validation, metadata management, and governance to maintain consistency.
Communication with Non-Technical Stakeholders	Medium severity because if the management or the non-technical decision-makers didn't understand the insights or the patterns properly, it will affect the suitability of the decision. This can reduce the business value of data analysis.	Effective and interactive communication affects the quality of the decision because it will rely on patterns determined by real datasets. Transforming technical information into non-technical information may delete some important details because it can not be interpreted. An alternative perspective is that data storytelling can communicate complex findings while keeping the main message clear with all the needed details	This issue has a significant organisational impact because the analysis on real organisational data and the insights should be clear to all management and every team member, which will enhance every process they produce. Recommends data storytelling to communicate analytical findings and visualisations effectively.
Undefined KPIs and Business Objectives	If the organisation doesn't focus on specific, clear objectives, it will be hard to measure how useful the results are and what techniques should be applied. It can also create unrealistic expectations for data specialists.	Clear KPIs work effectively to define the needed changes, provide measurable targets, and help evaluate analytical performance. KPIs depend on the business objective, what will be difficult when the objective is not clear yet. An alternative perspective is that KPIs should be reviewed and adapted as organisational priorities change.	Clear KPIs and business objectives are essential for successful data-driven decision-making because the analysis model used can be measured by comparing the results with the needed results. Organisations should establish measurable analytical and business KPIs before evaluating results. Also recommends well-defined metrics and business KPIs.

Overall Evaluation

Overall, **data security and privacy, data quality, and bias** have the highest impact because they directly affect the insights and results that impact decision-making, which may not be relatable to the organisation's problems. Other issues, such as communication, multiple data sources, and unclear KPIs, also may affect on the decision quality, including details and data quality, because of duplicates and

inconsistencies. Collecting data and only focusing on its quantity should not be the main target; the data should be clean, useful for analysis models, no duplicates, protect sensitive information, and reduce bias. This balanced approach allows the benefits of a data-driven culture to be achieved while reducing its major organisational risks.

Long-term impacts of using complex datasets for decision-making

Long-term usage of complex databases is projected to affect both organizations and end-users. To users, analyzing large amounts of variable data will provide opportunities for personalized services, enhanced user experience, and product/service recommendations based on their specific needs. At the same time, there could be increased risks of privacy and security breaches due to the increased amount of data being collected and analyzed, especially when sensitive information is involved. Organizations utilizing complex databases will have the ability to make more informed decisions with higher levels of accuracy through the utilization of data based on customer behavior, operational performance, and market trends. As such, over time organizations should be able to allocate resources in a more productive manner, operate at a higher level of efficiency, develop and implement strategies with higher degrees of effectiveness, and enhance their competitive position. Utilizing data continuously may also promote the growth of a data-driven culture within organizations where managers and employees become increasingly reliant on data as opposed to assumptions when making decisions.

Although the potential for positive outcomes exists as a result of using complex databases, the overall long-term outcome will be dependent upon the organization's ability to manage those databases. Managing large volumes of diverse data, which often originates from multiple sources and/or forms, can be time-consuming and labor-intensive. If the data contained within complex databases is inaccurate, incomplete, or inconsistent, it is possible that the results derived from analyses may be unreliable and potentially cause organizations to make uninformed decisions. Due to the fact that organizations collect sensitive information, protecting that information from unauthorized access and cyber-attacks is paramount. Additionally, if the data used for analyses contains biases or if the methodologies employed for analyses are suboptimal, it is possible that organizations may draw conclusions that are either unfair or inaccurate. Therefore, organizations employing complex databases will require effective data governance policies and procedures, data quality assurance programs, data security measures, as well as sufficient analytical capabilities to optimize the utilization of those databases.

In summary, the long-term implications associated with the use of complex databases will likely have a positive effect on users and organizations when those databases contain high-quality data that is stored securely and managed effectively. Users will have opportunities to receive more customized services in a more timely manner. Organizations will have the opportunity to make more accurate decisions and will be able to operate more efficiently, ultimately creating a competitive advantage. However, without appropriate data quality, privacy, security, and governance measures, the increasing complexity of datasets may create significant risks and reduce the reliability of decision-making.

Activity 2

The selected use case focuses on TV Show Data Analytics using the TVmaze API. TVmaze contains a large amount of data that performs information about shows and episodes, including genres, ratings, runtime, countries, networks, premiere dates, air dates, and schedules. This use case aims to enhance TVmaze's performance by using this data to determine relationships and convert the raw, complex data into meaningful insights that support the decision making process. Our use case focuses on analyzing and detecting patterns using weights and genres to check what kind of shows perform better for the target audience and get high ratings. This makes the use case suitable for demonstrating how Big Data analytics can support evidence-based decision-making.

Data-Driven Decision-Making Process

Data-driven decision-making (DDDM) is an enhanced version of the decision making process that depends on actual data from the organization's everyday services and produces insights used to take appropriate decisions that improve the organization's performance. The DDDM is not guided by existing data and makes decisions manually without a technical and management process; it requires a systematic process for collecting, analysing patterns and trends, and implementing decisions based on data.

1. Define Your Goal

The first stage is to identify the purpose of the use case study and what value you want to add to the organization by implementing this process. The goal should be clear and achievable for all the team members. **For the TV Show Data Analytics use case, the targeted goal is to understand the patterns of the high rating-low rating shows and relationships for high-weight and genre type shows to detect customer preferences and provide a customized experience.** The analysis focuses on the most influential data, such as weight, genre, runtime, country, and network. Focusing on a defined, specific goal makes the analysis results more accurate because of depending on useful information for the decision.

2. Establish a Hypothesis

The second stage involves establishing a hypothesis, which includes simple initial planning to identify the proposed way and method that will be used to achieve the goal. The hypothesis should study all the use case stages and parts to understand and detect where the best data could be used to have the greatest impact on TVmaze. After understanding the organizational areas, a suitable hypothesis could be: "TV shows have a certain relationship or specific pattern with genre and other characteristics that achieve higher ratings than others." Having an accurate, suitable hypothesis provides clear direction for the analysis.

3. Identify Data Need

The third stage plays a big role in the analysis result because it focuses on the needed data to achieve the business goal. The use case dataset will be obtained from the TVmaze API and will include quantitative and qualitative data. Qualitative data include genre, country, network, and show name, while quantitative information can include weight, runtime, and episode-related numerical values. The dataset has powerful side on the analysis because it contains date and time data that can be used to obtain a trend through time. Identifying these data requirements ensures that the collected information is sufficient and will produce high accuracy results.

4. Build Data Process

The fourth stage performs the needed process after identifying the needed specific data, so this process is to establish how the required data will be collected and prepared. The collected data should be reliable based on the source, and it should be clean to avoid wrong results because of inconsistency. This use case has a main, reliable data source. The data can be accessed through the accessible and related API, then loaded into the analysis environment using Polars because it is very large data. The data will then be checked for missing values, duplicates, inconsistent formats, and unsuitable records before analysis to produce reliable insights.

5. Analyze Data

The fifth stage is to check the suitable tool for the use case and prepare the needed techniques to perform with high accuracy. For the TVmaze API, polars perform the best tool for data manipulation and perform statistical analysis because its efficient for large scale data. Descriptive analysis can be performed to identify patterns and investigate genre, rating, and other characteristics, while visualisations such as bar charts and line charts can make the results easier to interpret using Plotly to develop interactive dashboards. The analysis can then determine whether particular genres or show characteristics are associated with higher ratings and identify important trends within the dataset.

6. Make a Decision

The final stage of using and analyzing the selected data is to use the insights that are obtained from the analytical techniques to make appropriate decisions that enhance the customers' experience and improve the organization's performance by providing customized services. That's why the insights should be performed on a way that both technical and the non technical can understand. **TV show characteristics, genres, and popularity weights are associated with higher ratings**, to considered the factors that make a show high rating show, then used to make content recommendations and plan content priorities. Then, use the evidence to support decisions about which types of content should receive greater consideration.

Decision: Give greater consideration to TV content with characteristics associated with high ratings, especially successful genres and higher-weight shows, for recommendations and content priorities.

Main Goal of the Selected Use Case

The main goal of applying advanced data analytics on the TVmaze dataset is to select the features that have a relationship with making the show more popular has more views, and to identify the patterns or trends these features have to provide insights from these complex data, then use the insights so the management can make decisions based on evidence. The input features must be genres, weight, networks, countries, runtime, and release information because these characteristics influence the show's popularity and customer demand. The generated insights can help decision-makers to use the large amount of stored data to determine which television content performs better based on different features, which patterns lead to high demand, and make evidence-based decisions

Organisation Description

The organisation selected for this case study is TVmaze. This organisation specializes in the digital media sector and is also operating and coordinating with television. TVmaze provides sufficient information about the organization in an organized, suitable, and accessible format for users, containing detailed data such as show titles, genres, ratings, networks, countries, air dates, schedules, and episode information all of these about television shows and episodes. The nature of this organisation's work is to collect all the company's essential information and providing them through a platform and API, which makes it accessible and can be used for analysis and other digital services. This makes TVmaze a suitable organisational context for the project because its large and varied datasets provide an opportunity to apply Big Data techniques to discover useful patterns and insights about television content.

Types, Sources and Volume of Data

The case study will involve structured and semi-structured data on television shows and episodes. The types of data that will be used include categorical data, such as genre, country, language, and network; numeric data, such as weight, running time, and data associated with episodes; textual data, such as names of shows and their descriptions; and date and time data, such as premiere date, airing date of an episode, and schedule. The major source of the data will be the TVmaze API, which contains information about TV shows, episodes, schedule, rating, genre, network, and country. The TVmaze API endpoints can be accessed and integrated to create a more comprehensive dataset for analysis. The required volume will include a huge

number of records obtained from several TVmaze endpoints, specifically from show and episode information, rather than a sample of data. The final volume will be determined after collecting all available data, and further processing of the dataset using Big Data techniques will be performed.

2.2 Comparison of Suitable Industry-Leading Tools and Software

For our selected use case, the TVmaze API and many software tools would work effectively to turn the complex data into useful insights. Here are tools to handle large-scale and complex data

Tool / Software	Key Capabilities	Strengths	Limitations
Apache Spark (PySpark)	This tool can handle large scale data and clean it of missing values, duplicates, and inconsistencies, and aggregate it into statistical groups, which makes it easier to understand to chose how to analyze it to figure out information from a complex dataset	Process large datasets effectively with different machine learning models and can process various data sources without complexities	Machine learning models need much more processing and handling to generate effective results; requires more computational resources and is complex for beginners
Polars	High performance for large data preprocessing, including cleaning, scaling, transformations, filtering, and aggregations	Performs effectively for fast operations; Python-easy to use to generate effective results; performs very well with no complexities for large data because of the large memory to use and store, so it works effectively for TV maze with 90k rows	Focuses on dataframe manipulation more than large data pipeline. Does not provide the same extensive machine-learning ecosystem as Spark.
Microsoft Power BI	Provides different charts, figures, KPIs, and filtering to end with an interactive dashboard for complex data and can apply different groupby stages to different data	Easy to use; creates an easy-to-interact-with dashboard for technical and non-technical decision makers; can display different forms of data	Some advanced features require paid/licensed versions; very large datasets may require careful data modelling and become more difficult to use and require knowledge of DAX .
Plotly	Can transform different data in different forms into interactive dashboards and charts through Python	Integrates with Python analysis workflow, highly customizable, works within easy-to-access Python workplaces	Primarily focused on visualisation rather than large-scale data processing; advanced dashboard development can require additional programming knowledge.
n8n	It performs automation workflows, takes API data from Url provide data transformation and filtering on the uploaded data, and connects different applications and services.	It provides direct connections to APIs from the HTTP Request node; it works effectively with JavaScript and Python, and reduces waiting time to implement a repetitive process	To implement advanced automation work flow implements programming knowledge and experience; loading big data may take some time. It is mainly an automation and integration tool rather than a dedicated Big Data processing it doesn't work like an ML tool, and workflows may require careful resource management.
Python Data Pipeline	It works effectively as a sequence datatasks for a data set all of transformation, cleaning, and analysis	It's easy to change to match the targeted requirements and can connect multiple python libraries to work together,	It requires programming knowledge to implement advanced workflows, errors through piplenes stops the

	also it stores the data effectively through scripts	can work effectively with Python tools and the results can be saved through csv file such as Polars, PySpark, and Plotly.	whole excutions , and complex pipelines can become difficult to manage and maintain compared with visual automation tools.
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Selected Tools and Software

After researching all the big data handling tools and understanding the differences between them to finally choose the best tool to perform effective data manipulation, transformation, storage, applying machine learning techniques, and interactive dashboards, **PySpark, Polars, and Plotly** were selected for the large scale data TVmaze use case.

PySpark is a very effective tool for the analysis phase because it offers Spark MLlib for machine learning. is appropriate for processing the dataset because it can handle large datasets and perform different transformations. **Polars** was selected for collecting the data from its source, cleaning, transforming, filtering, and aggregating the TVmaze data because it provides fast DataFrame operations and is easy to use within Python. **Plotly** was selected for creating the interactive dashboards because it integrates directly with Python for preprocessing and analysis and allows different types of charts and interactive visualisations to be created from the processed data to understand the complex data and turn it into insights. **n8n** was selected to perform automated workflows on large scale data collected from the TVmaze API, which worked effectively in downloading the data without very long waiting times or any errors in the data values. It was used to connect directly to the API through an HTTP Request to load it and use it through execution, retrieve multiple pages of TV show data, clean and transform the collected records to perform classification correctly and in aggregated groups of data, classify shows as High Rated or Low Rated, and convert the final results into a CSV file that could be read and written. Using n8n reduces the need to manually repeat these steps whenever new data is collected and creates a clear automated workflow from API data collection to the final prepared file. All these steps didn't take much time; it automated the whole workflow effectively and quickly, but a Python data pipeline does **not** provide a visual workflow interface like n8n.

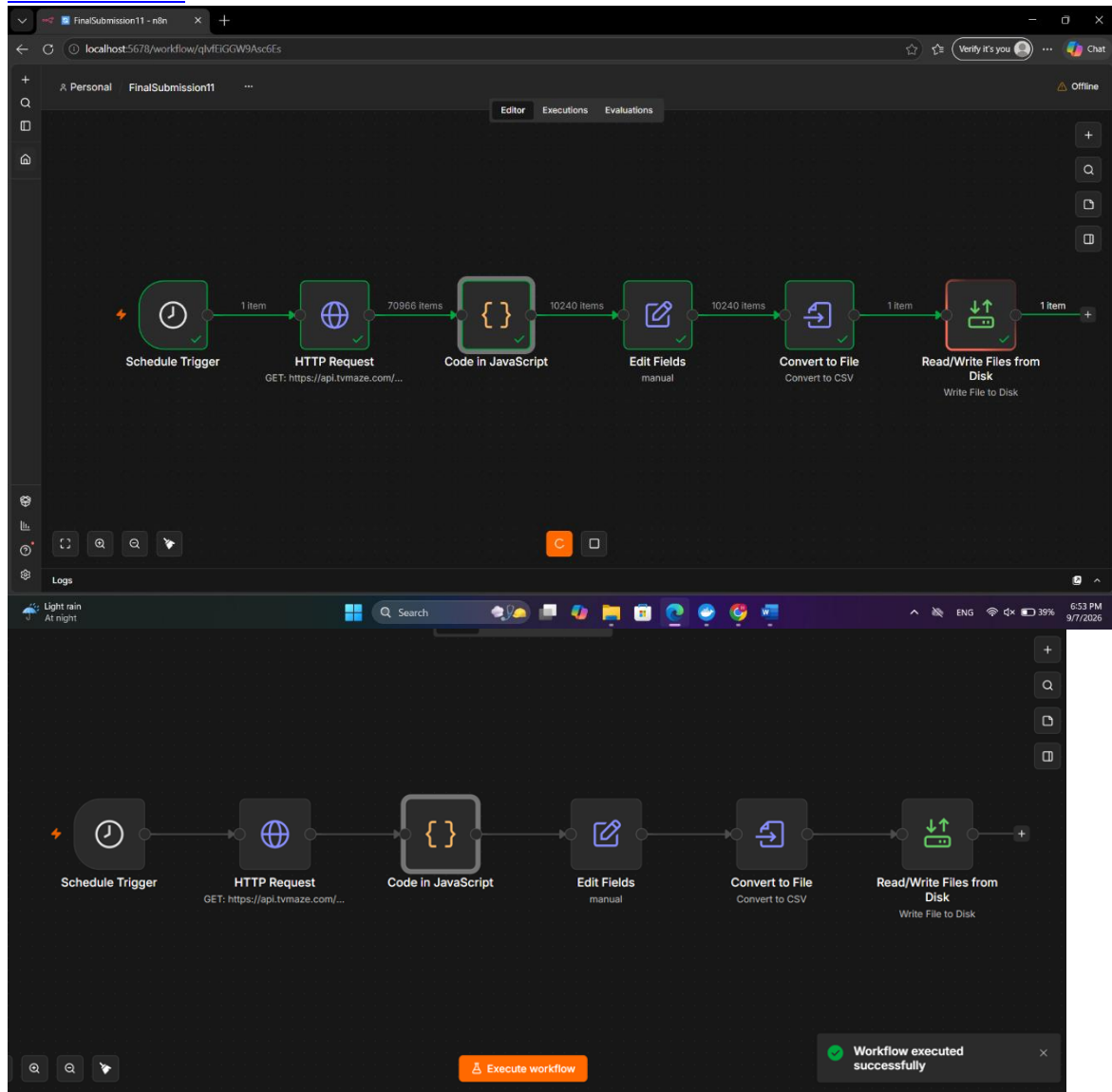
These tools were selected to perform preprocessing, analysis, and dashboards for TVmaze API data in an all-Python workflow; it all worked together to process a complex, large dataset. **Power BI** also provide interactive dashboards for data, but Plotly was chosen because it can be used directly in the Python environment after the preprocessing and classification because it will make the charts and dashboards clearer also it Works well with DataFrames such as Polars and gives more control over the visualisations, but powerBi Very large datasets may require careful data modelling which makes it limited for the TVmaze API. Similarly, Polars is more straightforward for the data preparation stage, while PySpark provides stronger capabilities when machine learning and larger-scale processing are required with large memory processing; it works effectively for complex analytical workflows, and n8n handles pagination without manually collecting each page.

It can perform the needed processing before classification, clean inconenssity values, duplicates, and missing values, which allows the classification to run faster and be more consistent. Therefore, the selected combination provides a good balance between **data preparation, Big Data processing, machine learning, and interactive visualisation**, making it suitable for analysing the TVmaze dataset and supporting data-driven decisions.

evidence

- **Apache Software Foundation (2026) MLlib: Main Guide.** This documentation explains how PySpark effectively performs services for data preprocessing, cleaning, transformation, scaling, and types of analysis, regression and classification, and vector assembly. It supports the use of PySpark for the machine-learning stage of the TVmaze project. Available at: [Apache Spark MLlib Documentation](#)
- **Polars (2026) Polars User Guide.** This documentation describes Polars as a very effective library for loading big data and applying its capabilities in high performance workflow for data manipulation, filtering, transformation, aggregation, and query optimisation. It supports the selection of Polars for preparing and cleaning the TVmaze dataset. Available at: [Polars Documentation](#)

- **Plotly (2026)** *Plotly Python Documentation*. This documentation explains how Plotly performs complex data effectively in all the needed types of charts and figures in full interactive dashboards to support insights and decisions making. It provides evidence for using Plotly to create the interactive dashboards in the project. Available at: [Plotly Python Documentation](#) .
- **n8n (2026)** *HTTP Request Node Documentation*. This documentation explains how n8n can connect to APIs using HTTP requests, call the data to use it, apply JavaScript, and perform analysis to implement automated workflows. It supports the use of n8n for automatically collecting and processing data from the TVmaze API. Available at: [n8n HTTP Request Documentation](#) .



Implimentaion

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Page 349: 242 records | Total: 83153
Page 350: 240 records | Total: 83393
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Page 351: 243 records | Total: 83636
Page 352: 247 records | Total: 83883
Page 353: 247 records | Total: 84130
Page 354: 244 records | Total: 84374
Page 355: 244 records | Total: 84618
Page 356: 247 records | Total: 84865
Page 357: 246 records | Total: 85111
Page 358: 241 records | Total: 85352
Page 359: 243 records | Total: 85595
Page 360: 243 records | Total: 85838
Page 361: 249 records | Total: 86087
Page 362: 241 records | Total: 86328
Page 363: 247 records | Total: 86575
Page 364: 248 records | Total: 86823
Page 365: 247 records | Total: 87070
Page 366: 245 records | Total: 87315
Page 367: 245 records | Total: 87560
Page 368: 245 records | Total: 87805
Page 369: 246 records | Total: 88051
Page 370: 239 records | Total: 88290
Page 371: 248 records | Total: 88538
Page 372: 242 records | Total: 88780
Page 373: 247 records | Total: 89027
Page 374: 248 records | Total: 89275
Page 375: 249 records | Total: 89524
Page 376: 212 records | Total: 89736
No more pages. Data collection finished.

Final number of records: 89736
```

Collecting data from Tvmaze API

```
> print("Shape:", df.shape)
print("Columns:", df.columns)

'' Shape: (89736, 14)
Columns: ['id', 'name', 'type', 'language', 'genres', 'status', 'runtime', 'average_runtime', 'premiered', 'ended', 'rating', 'weight', 'network', 'country']

df.head(10)

shape (10, 14)
id name type language genres status runtime average_runtime premiered ended rating weight network country
164 str str str str str str float64 float64 str str float64 float64 str str
1 "Under the Dome" "Scripted" "English" "Drama, Science-Fiction, Thrill..." "Ended" 60 60 "2013-06-24" "2015-09-10" 6.6 100 "CBS" "United States"
2 "Person of Interest" "Scripted" "English" "Action, Crime, Science-Fiction" "Ended" 60 60 "2011-09-22" "2016-06-21" 8.8 100 "CBS" "United States"
3 "Bitten" "Scripted" "English" "Drama, Horror, Romance" "Ended" 60 60 "2014-01-11" "2016-04-15" 7.4 99 "CTV Sci-Fi Channel" "Canada"
4 "Arrow" "Scripted" "English" "Drama, Action, Science-Fiction" "Ended" 60 60 "2012-10-10" "2020-01-28" 7.4 99 "The CW" "United States"
5 "True Detective" "Scripted" "English" "Drama, Crime, Thriller" "Running" 60 63 "2014-01-12" null 8.1 100 "HBO" "United States"
6 "The 100" "Scripted" "English" "Action, Adventure, Science-Fic..." "Ended" 60 60 "2014-03-19" "2020-09-30" 7.7 100 "The CW" "United States"
7 "Homeland" "Scripted" "English" "Drama, Thriller, Espionage" "Ended" 60 60 "2011-10-02" "2020-04-26" 8.2 99 "Paramount+ with Showtime" "United States"
8 "Glee" "Scripted" "English" "Drama, Music, Romance" "Ended" 60 60 "2009-05-19" "2015-03-20" 6.6 98 "FOX" "United States"
9 "Revenge" "Scripted" "English" "Drama, Thriller, Mystery" "Ended" 60 60 "2011-09-21" "2015-05-10" 7.8 97 "ABC" "United States"
10 "Grimm" "Scripted" "English" "Drama, Crime, Supernatural" "Ended" 60 60 "2011-10-28" "2017-03-31" 8.4 99 "NBC" "United States"
```

```
df.null_count()

shape: (1, 14)
id name type language genres status runtime average_runtime premiered ended rating weight network country
u32 u32 u32 u32 u32 u32 u32 u32 u32 u32 u32 u32 u32 u32
0 0 0 2803 0 0 25144 9366 3459 22609 78010 0 23927 23927

print("Rows before removing duplicates:", df.height)

df = df.unique(subset=["id"])

print("Rows after removing duplicates:", df.height)

Rows before removing duplicates: 89736
Rows after removing duplicates: 89736
```

TAKE AN OVERALL SUMMARY OF THE COLUMN TYPES AND DATASET SHAPE

Data type conversion

```
df = df.with_columns([
    pl.col("premiered").str.to_date(strict=False),
    pl.col("ended").str.to_date(strict=False)
])
```

df.schema

```
Schema([('id', Int64),
        ('name', String),
        ('type', String),
        ('language', String),
        ('genres', String),
        ('status', String),
        ('runtime', Int64),
        ('average_runtime', Int64),
        ('premiered', Date),
        ('ended', Date),
        ('rating', Float64),
        ('weight', Int64),
        ('network', String),
        ('country', String)])
```

```
df.select([
    pl.col("rating").is_null().sum().alias("missing_rating"),
    pl.col("runtime").is_null().sum().alias("missing_runtime"),
    pl.col("average_runtime").is_null().sum().alias("missing_average_runtime")
])
```

shape: (1, 3)

missing_rating	missing_runtime	missing_average_runtime
u32	u32	u32
78010	25144	9366

Scaling the data till it became easy to use

```
Data Cleaning
1 df_clean = df.filter(pl.col("rating").is_not_null())
2
3 print("Original rows:", df.height)
4 print("Clean analysis rows:", df_clean.height)
5
6 Original rows: 88736
7 Clean analysis rows: 11730
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9 df_clean.head(10)
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11 shape: (10, 14)
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```

```
genre_df = (df_clean.with_columns(pl.col("genres").str.split(", ").explode("genres"))
genre_counts = (genre_df.group_by("genres").agg(pl.lit().alias("Number of Shows")).sort("Number of Shows", descending=True))
genre_counts

***
shape (28, 2)
  genres Number of Shows
  str          u32
  "Drama"      4922
  "Comedy"     3470
  "Crime"      1855
  "Action"     1678
  "Romance"    1382
  ...
  "War"        133
  "Espionage"  110
  "Western"    72
  "Adult"      58
  "DIY"        55

genre_counts_clean = genre_counts.filter(pl.col("genres") != "")
genre_counts_clean

***
shape (28, 2)
  genres Number of Shows
  str          u32
  "Drama"      4922
  "Comedy"     3470
  "Crime"      1855
  "Action"     1678
  "Romance"    1382
  ...
  "War"        133
  "Espionage"  110
  "Western"    72
  "Adult"      58
  "DIY"        55
```

Data cleaning

```
df_clean = df_clean.with_columns(pl.col("runtime").fill_null(pl.col("average_runtime").alias("runtime")))
df_clean.select([pl.col("runtime").is_null().or().alias("missing_runtime")])

***
shape (1, 1)
  missing_runtime
  u32
  57

df_clean = df_clean.with_columns(pl.col("premiered").dt.year().alias("premier_year"))
df_clean.select(["name", "premiered", "premier_year", "rating", "runtime"]).head(10)

***
shape (10, 5)
  name      premiered premier_year rating runtime
  str       date      f64      f64      f64
  "Schitt's Creek"  2015-01-13  2015  7.6  30
  "Dirty Business"  2028-02-23  2028  8.5  65
  "Strong Women Do Bang Soon"  2017-02-24  2017  8.0  60
  "Nightwing: The Series"  2014-09-29  2014  6.5  10
  "Moon Embracing the Sun"  2012-01-04  2012  7.4  70
  "The Sevenside"  2024-01-19  2024  9.2  47
  "Vivienne"  2023-05-12  2023  6.5  32
  "Staring 8"  2024-10-30  2024  6.5  48
  "Hot and Catch Fire"  2014-06-01  2014  8.1  60
  "90 Day Fiancé: Before the 90 D...  2017-08-08  2017  6.6  120

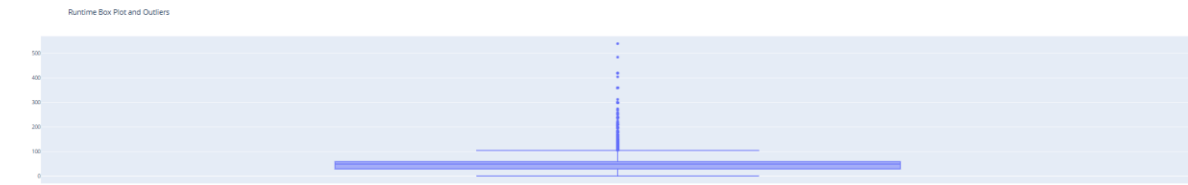
df_clean.describe()

***
shape (5, 16)
  statistic  id      name      type      language      genres      status      runtime      average_runtime      premiered      ended      rating      weight      network      country      premier_year
  str        f64      str       str       str       str       str       f64      f64      str       str       f64      f64      str       str       f64
  "count"    11725.0  ""      "11725"  "11725"  "11557"  "11725"  "11725"  11059.0  11051.0  "11723"  "9283"  11725.0  11725.0  "9190"  "9190"  11723.0
  "null_count"  0.0  "0"  "0"  "0"  "0"  "0"  "0"  57.0  125.0  "3"  "2433"  0.0  0.0  "3536"  "3536"  3.0
  "mean"    32874.215419  null  null  null  null  null  null  45.151058  45.173608  "2019-08-04 18:30:48.056370"  "2014-06-03 14:31:18.633530"  0.710549  80.528398  null  null  2013.107008
  "std"    26380.541024  null  null  null  null  null  null  22.678408  22.683818  null  1.13345  17.143394  null  null  12.842414
  "min"      1.0  "1"  "1"  "Happy Family USA"  "Animation"  "Afrikaans"  ""  "Ended"  1.0  1.0  "1925-05-01"  "1950-10-20"  1.0  0.0  "1-1"  "Argentina"  1926.0
  "25%"     6502.0  null  null  null  null  null  null  30.0  30.0  "2010-08-20"  "2012-03-22"  6.2  71.0  null  null  2010.0
  "50%"     23762.0  null  null  null  null  null  null  48.0  48.0  "2015-08-26"  "2016-07-15"  7.0  86.0  null  null  2017.0
  "75%"     42491.0  null  null  null  null  null  null  80.0  80.0  "2021-08-06"  "2022-01-14"  7.5  94.0  null  null  2021.0
  "max"     93648.0  "Xosha"  "Variety"  "Zulu"  "Western"  "To Be Determined"  219.0  219.0  "2025-06-04"  "2026-09-07"  9.8  100.0  "XO"  "United States"  2026.0

rating_stats = df_clean.select()
```

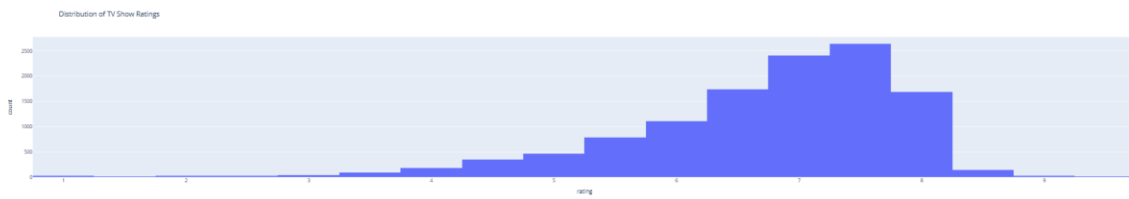
Filling the null values and take describe for statical values

```
fig = px.box(
    df.select(["runtime"]),drop_na(),
    y="runtime",
    title="Runtime Box Plot and Outliers"
)
fig.show()
```



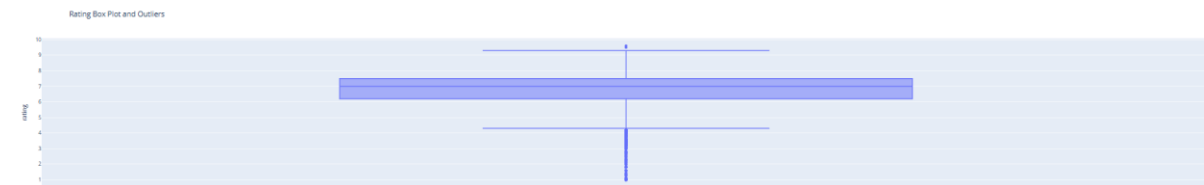
```
rating_data = (
    df
    .select(["rating"])
)
```

```
rating_data = (
    df.select(["rating"])
    .drop_na()
)
fig = px.histogram(
    rating_data,
    x="rating",
    nbins=30,
    title="Distribution of TV Show Ratings"
)
fig.show()
```



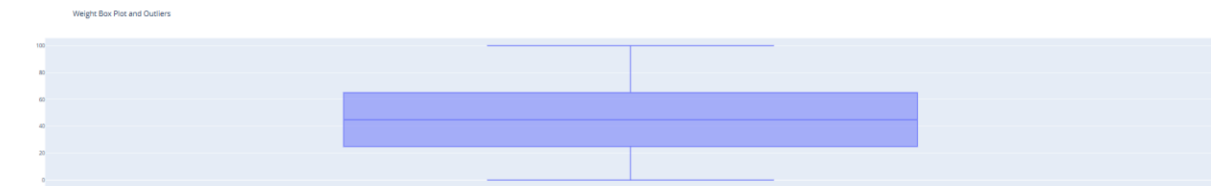
```
fig = px.box(
    df.select(["rating"]),drop_na(),
    y="rating",
    title="Rating Box Plot and Outliers"
)
fig.show()
```

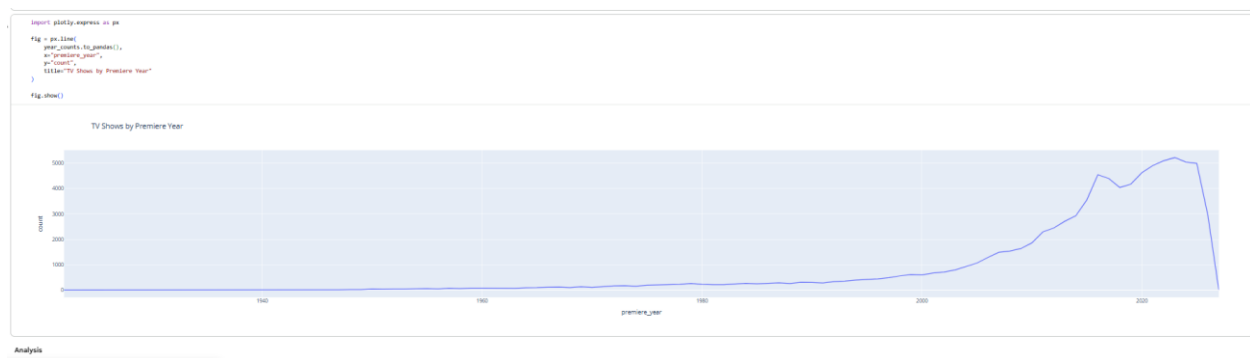
```
fig = px.box(
    df.select(["rating"]),drop_na(),
    y="rating",
    title="Rating Box Plot and Outliers"
)
fig.show()
```



```
fig = px.box(
    df.select(["weight"]),drop_na(),
    y="weight",
    title="Weight Box Plot and Outliers"
)
fig.show()
```

```
fig = px.box(
    df.select(["weight"]),drop_na(),
    y="weight",
    title="Weight Box Plot and Outliers"
)
fig.show()
```





Defining the outliers of the large-scale data set

```
ml_df["weight"].min(),
"-",
ml_df["weight"].max()
)

print(
    "Unique IDs:",
    ml_df["id"].n_unique()
)
```

Only 57769 unique shows are available with the current filters.

FINAL SHOW ML DATASET
Rows: 57769
Columns: 12
Runtime: 1.0 → 180.0
Premiere: 1990-01-01 → 2026-05-31
Rating: 1.0 → 9.5
Weight: 0.0 → 100.0
Unique IDs: 57769

** Total Shows: 57769

Low Rated (< 5):
50871

High Rated (>= 5):
6898

Sample:
shape: (10, 7)

name	genres	runtime	premiered	rating	weight	label
---	---	---	---	---	---	---
str	str	f64	date	f64	f64	i32
Jacht op Guido & Emma		5.0	2016-06-30	null	18.0	0
Baby Beauty Queens		30.0	2010-07-27	null	23.0	0
Синья роза	Drama	45.0	2017-08-06	null	28.0	0
World's Greatest Head to Head	Sports	30.0	2018-05-28	null	45.0	0
Aa, Love Hotel: Gokaban		29.0	2014-11-23	null	26.0	0
Raeng Tian	Drama, Romance	75.0	2019-09-02	null	44.0	0
Dead of Summer	Drama, Horror	60.0	2016-06-28	6.3	91.0	1
Vivre avec les volcans	Travel	52.0	2018-12-31	null	4.0	0
Морские дьяволы. Вектор атаки		45.0	2024-12-16	null	13.0	0
Faites comme chez vous !	Comedy	52.0	2005-04-03	null	9.0	0

Create a dataframe for the analysis method without outliers to create accurate results to support decision-making

```
print("\nBalanced Training Distribution:")
balanced_train_df.groupby("label").count().orderBy("label").show()

print("Balanced Training rows:", balanced_train_df.count())
```

```
*** Training rows: 3284
    Testing rows: 739
```

Original Training Distribution:

```
+-----+-----+
|label|count|
+-----+-----+
|  0 | 277 |
|  1 | 2927 |
+-----+-----+
```

Balanced Training Distribution:

```
+-----+-----+
|label|count|
+-----+-----+
|  0 | 2986 |
|  1 | 2927 |
+-----+-----+
```

Balanced Training rows: 5833

Training and blancing

```
print("CONFUSION MATRIX")

best_predictions.groupBy(
    "label",
    "prediction"
).count().orderBy(
    "label",
    "prediction"
).show()
```

```
*** BEST DECISION TREE - TEST RESULTS
```

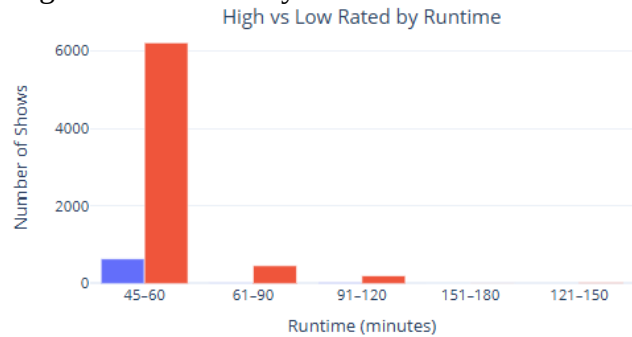
```
Accuracy : 81.87%
Precision : 88.31%
Recall : 81.87%
F1-Score : 84.75%
```

CONFUSION MATRIX

```
+-----+-----+-----+
|label|prediction|count|
+-----+-----+-----+
|  0 |      0.0 |   13 |
|  0 |      1.0 |   38 |
|  1 |      0.0 |   96 |
|  1 |      1.0 |  592 |
+-----+-----+-----+
```

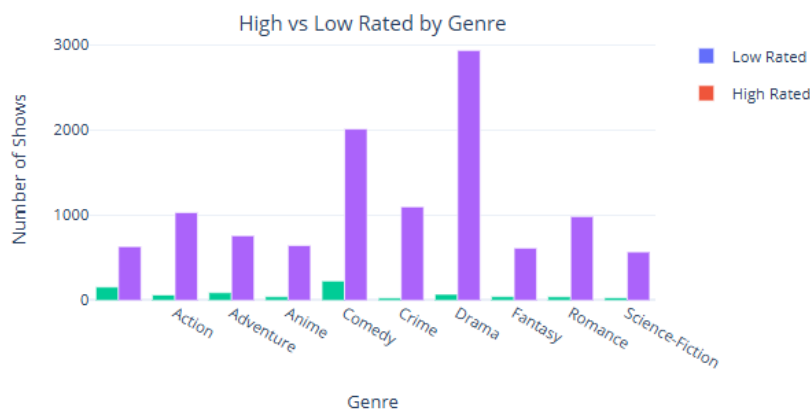
- **81.87% accuracy** means the model correctly classified about 82% of the TV shows as High Rated or Low Rated.
- The confusion matrix shows **592 High Rated shows were correctly classified**, while 38 were incorrectly classified as Low Rated; for Low Rated shows, 13 were correctly classified and 38 were incorrectly classified as High Rated.

High vs Low Rated by Runtime



In the current chart, we see that high versus low ratings are measured by show duration, and that most television programs run between 45 and 60 minutes, making this the most common timeframe for both high- and low-rated programs. However, the number of high-rated programs in this category is significantly higher than the number of low-rated programs. The 61-90 minute and 91-120 minute categories have considerably fewer programs. While the 121-150 minute and 151-180 minute categories contain very few programs, we see, as illustrated, that longer show durations are less prevalent in the information provided. Most available television programs follow an episode length, and the results indicate that the dominant show duration among the analyzed programs ranges between 45 and 60 minutes, thus being more common for highly rated content. In addition, there is no need to prove that the chart's display time leads to higher viewership, because the number of programs in each category is not equal. So, when planning content, companies can focus on programs that range in length from 45 to 60 minutes, taking into account other factors such as type and the program's reputation before finalizing.

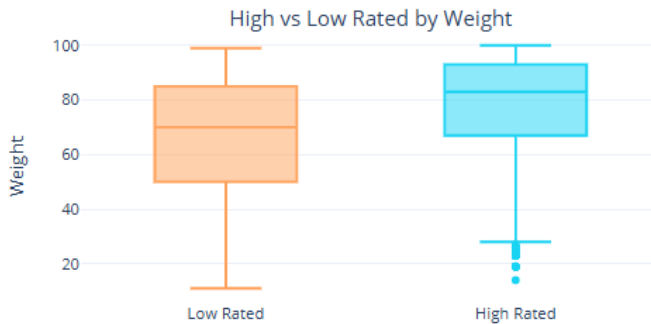
2. High vs Low Rated by Genre



The graph comparing high-rated and low-rated shows by different genre illustrates that there are the most Drama programs, followed by Comedy, Crime, Action, and Romance genres. The genres of Science-Fiction, Fantasy, and Adventure have less number of shows produced. Among the shown genres, the number of high-rated shows exceeds the number of low-rated shows, meaning that the dataset contains more shows qualified as highly rated considering rating threshold of five. The big number of Drama and Comedy shows indicates that these genres are widely represented in TVmaze, which makes them valuable for further analysis. However, the number of shows in a genre shouldn't be seen as a basis for concluding that the given genre has better rating, as having more shows means a higher number of rated high and low programs. Thus, the matter is helpful in determining the popularity and representation of genres, while the average rating investigation is the supplementary evidence on the performance of ratings. With reference to the stated purposes of use, the findings may assist in content recommendations due to

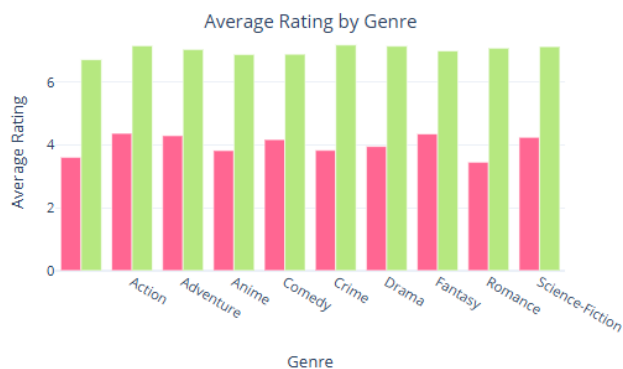
determination of the genres with rich representation and should be taken into account when examining successful TV content.

3. High vs Low Rated by Weight



The box plot that compares high ratings and low ratings by means of weight shows one important aspect about the popularity weight between the high-rated shows and the low-rated shows. The median popularity weight for high-rated shows is around 82, while for low-rated shows, it is around 70. Therefore, it can be established that the high-rated shows have higher TVmaze popularity weights. Furthermore, the high-rated shows show a relatively high concentration of observations about the weight, while low-rated shows show a wider distribution of weight which confirms the positive correlation between weight and rating classification. Despite the presence of some low weight cases within the category of high-rated shows, the data indicate that not all shows require a high weight in order to get a high rating. At the same time, the overlap shows that weight alone cannot be used in order to make conclusions about shows' high ratings or low scores. In this connection, the results suggest that weight shall be complemented by other characteristics like genre, duration and year of the first release.

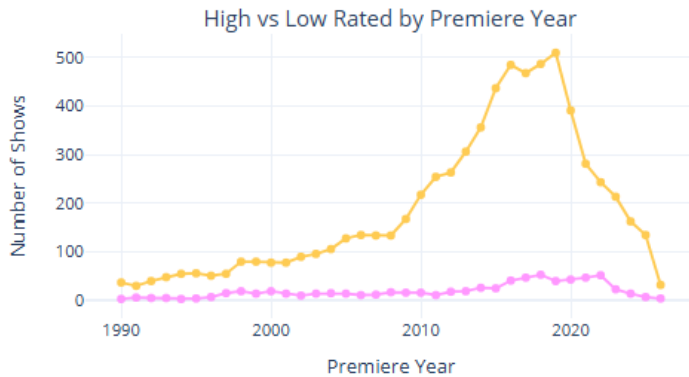
4. Average Rating by Genre



The Average Rating by Genre graph illustrates that there is a significant difference between the mean ratings of high rated TV series and low rated TV series in those genres. The mean rating of high rated series stays approximately at 7 and becomes even higher in the majority of cases, while the mean rating of low rated series is significantly lower, averaging between about 3.5 and 4.5. In the high rated group, there are a number of genres that indicate a high average rating, like Action, Adventure, Crime, Drama, and Fantasy, while Romance rates relatively low among high rated genres. The low rated group differs among genres as well, since Action, Adventure, and Sci-Fi rate a bit high compared to other low rated genres. These differences imply that rating results are not the same for every genre, while genre itself can give valuable insights when looking at ratings data. Still, this evidence

should not be seen as a proof of the meaning that a certain genre causes higher ratings, since other factors may also contribute to the audience ratings. In conclusion, the chart provides enough evidence to identify genres which perform better from the rating point of view and can help make

5. High vs Low Rated by Premiere Year



The High vs Low Rated by Premiere Year chart illustrates the variation in the number of high-rated and low-rated shows over time. The number of high-rated shows rises continuously from the 1990s, with a sharp increase around the year 2010, reaching its peak near the year 2019. After this year, the number of high-rated shows drops sharply in early 2020s. The low-rated group remains much smaller in any given year; however, it also sees some increase in the later years. Such trends may signal that there is much more data represented in the dataset in the modern time period, especially in late 2010s. The decrease after the year 2019 cannot be treated as a sign of dropping popularity or quality of TV shows as this is likely to be connected with the number of shows present in the given period of time. In the context of this study, it is also worth mentioning that the documentary shows have been produced in different proportions over the years and thus should be analyzed considering the premiere year when speaking about ratings.

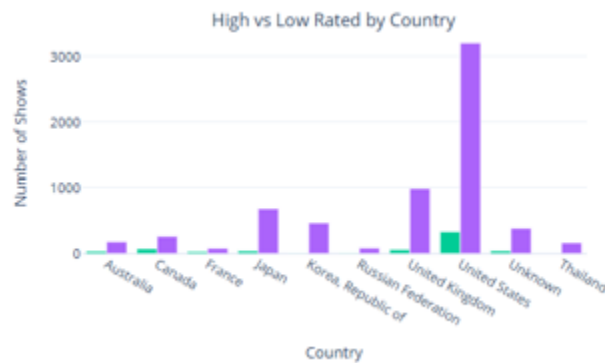
In general, the first data panel shows that show duration, genre, and year of airing are all useful characteristics for understanding the differences between highly rated and low-rated television programs. It highlights the strongest relationship between viewership and rating, with highly rated programs enjoying higher acceptance rates. The data panel also shows that show durations between 45 and 60 minutes are the most important category, and that drama and comedy are among the most requested genres. Because the dashboard shows that show durations between 45 and 60 minutes are the most popular and preferred, and that drama and comedy are among the most requested genres, these results provide evidence supporting the project's final decision: greater interest in showcasing these features associated with higher ratings, and when a feature is present alone, it receives a higher rating.

1. High vs Low Rated by Language



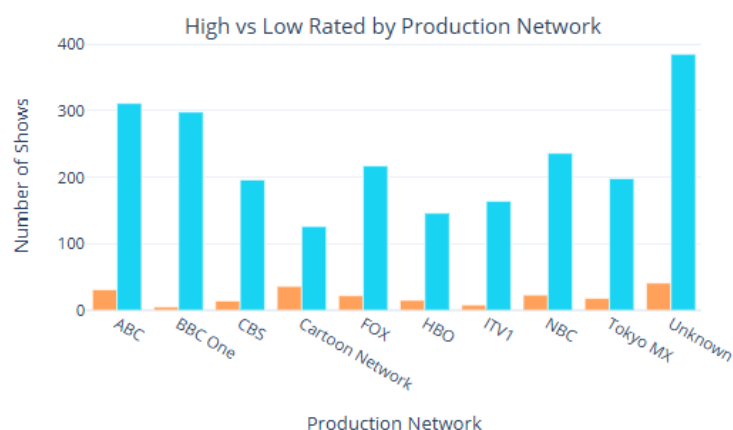
The chart showing High Rated vs Low Rated by Language highlights the fact that English tops the chart, with many more high-rated and low-rated shows than those in any other language. Other languages in line with it are Japanese and French, whereas there are fewer shows in languages such as German, Korean, Russian, Spanish, Thai, and Turkish. English's supremacy tells us that the dataset leans towards English-language shows and this probably is due to the high volume of English-language content available in TVmaze's dataset. It can also be said about the high-rated English shows that their number is really impressive even in comparison to the number of low-rated English shows. Still, that doesn't mean that English shows are really better rated, as the number is high for this language. Hence, language could be useful in understanding where the market and audience of TV content are located, but it needs to be used together with other indicators. Organizations could use it in the framework of the case and identify, where the major markets are, without jumping to conclusions only based on the number of shows.

2. High vs Low Rated by Country



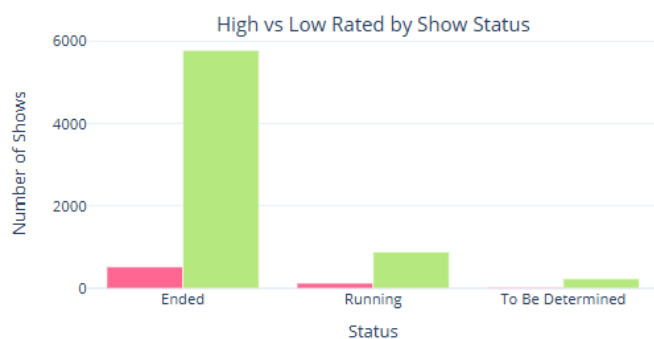
According to the graphic shown on High and Low Rated by Country, the US has the highest number of samples included in this chart, with the UK in second place and Japan in third. It is important to note that the US has an exceptionally high number of shows in comparison to other countries. Other countries like Japan, the UK, or South Korea are also well represented in this sample, albeit not in the same extent. This clearly indicates that this chart has a large concentration of TV productions from prominent global markets. However, having more shows does not necessarily mean that this country has better quality productions, since this fact heavily depends on the quantity of all TV shows produced in each country.

3. High vs Low Rated by Production Network



High vs Low Rated by Production Network is a chart that illustrates the differences in terms of high and low-rated shows linked to each production network. Among the networks shown, Unknown is the most represented in terms of number of shows, followed by ABC and BBC One with lesser numbers recorded for Cartoon Network, CBS, HBO and TV Tokyo. Most networks have more low-rated shows than high-rated ones. The Unknown category is especially valuable since it represents the shows that have not had their production network included in the dataset. This limits the application of network information for decision-making since it denies a possibility of making accurate comparison of production companies or broadcasters. Still, this chart is helpful for determining the representation of networks in the dataset. For the case in question, production network can be taken as a complementary item in the analysis of successful TV content.

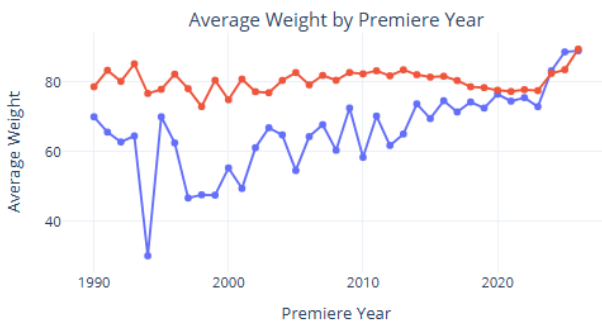
High vs Low Rated by Show Status



The High vs Low Rated show status chart illustrates that the status of Ended shows is the most prominent one in the series analyzed with only Running series being in the second position while To Be Determined having only a limited number of series. The abundance of ended series indicates that in the data analyzed the number of completed shows is much bigger than that of running shows. The shows that ended feature a significant amount of high-rated shows vs low-rated shows, while the same tendency should also be valid for running shows. The fact that the concept of rating classification is applicable to different statuses of TV shows apart from ongoing programs weakens the potential for direct comparison between the statuses due to the significant variation in quantity. The resultant finding

may be applicable in TV content distribution and, consequently, assist in differentiating between the types of shows that people prefer.

5. Average Weight by Premiere Year



The High vs Low Rated show status chart illustrates that the status of Ended shows is the most prominent one in the series analyzed with only Running series being in the second position while To Be Determined having only a limited number of series. The abundance of ended series indicates that in the data analyzed the number of completed shows is much bigger than that of running shows. The shows that ended feature a significant amount of high-rated shows vs low-rated shows, while the same tendency should also be valid for running shows. The fact that the concept of rating classification is applicable to different statuses of TV shows apart from ongoing programs weakens the potential for direct comparison between the statuses due to the significant variation in quantity. The resultant finding may be applicable in TV content distribution and, consequently, assist in differentiating between the types of shows that people prefer.

Overall, Dashboard 2 shows that language, country, production network, show status and popularity weight provide useful market and production information about TV shows. The dataset is strongly concentrated around English-language content and major production markets such as the United States and United Kingdom. The production-network analysis also highlights the importance of handling missing information, particularly the large Unknown category. Now we come to the other clear figure. The second information panel shows that language, country, production network, television program status, and its acceptance rate among peoples make it easier for us to provide important information about program production and its television subject matter. This information is largely concentrated data about English-language content and major production markets such as the United States and the United Kingdom, as is now shown in the analysis of the production network, which is important in addressing the missing information.

Evaluate the data preparation and manipulation methods

After the company completed the process, it was able to use a more effective dataset to discover trends and create interactive graphs. The comprehensive data cleaning and transformation engineering laid a solid foundation, thus minimizing potential problems related to low-quality or inappropriate data for children or specific demographics, which could have affected the final results. This allowed for the appropriate processing of the classification field by the machine learning model, and the data was filtered to include only entries within the time frame to ensure the analyzed entries were relevant to the study's scope. In addition, cleaning, organizing, and standardizing the data makes the results more accurate and reliable for making decisions that lead to the company's best interests. Descriptive statistics and outliers were reviewed to determine how different numerical fields are distributed in the study population. With these processes complete, the data set was now able to be utilized more effectively to discover trends,

generate interactive visuals, and apply the decision tree classification model. As such, the overall data cleaning process and feature engineering/transformations established a solid foundation for the remainder of the study and reduced potential issues stemming from low-quality or inappropriate data impacting the eventual conclusions. This allowed the categorical genre field to be processed appropriately by the machine learning model. Finally, the data was filtered to include only entries within a certain date range and/or runtime range to ensure that the entries analyzed fit the scope of the study. Additionally, descriptive statistics and outliers were reviewed to determine how various numerical fields are distributed throughout the study population, and to find any unusual values. Overall, these processes were successful because they enhanced the quality and organization of the data set while retaining valuable data for analysis purposes. With these processes complete, the data set was now able to be utilized more effectively to discover trends, generate interactive visuals, and apply the decision tree classification model. As such, the overall data cleaning process and feature engineering/transformations established a solid foundation for the remainder of the study and reduced potential issues stemming from low-quality or inappropriate data impacting the eventual conclusions.

Evaluation of Tools and Software Used

During the development of the analysis of the tvmaze api classification to generate results and insights that benefit and simplify the decision making process. **n8n** performed the main tool to apply automated classification for the tvmaze api. Its main strength for the decision-making process was reducing manual work by automating API requests and loading large-scale data, pagination, filtering, transformation of different data values, and CSV file generation easily without entering with accurate results. However, its disadvantage that it wasn't flexible with applying machine learning models to reduce the needed steps to apply the need process that will be generated at python. **Polars** was used for data preparation and manipulation. Its strengths was effectively simple syntax to clean large data and understand the syntax of it while cleaning duplicates and missing values. Its main weakness is that it requires programming knowledge and still doesn't provide machine learning analysis services its only efficient in tv maze API at cleaning and transforming. **PySpark** was used for Big Data analysis and machine learning, particularly the Decision Tree classification model that performed effectively to classify the high rating and low rating shows. Its major advantage is its ability to process large datasets and provide scalable machine learning capabilities through MLlib without complexities to apply the machine learning services. However, PySpark can be more complex to use and may require more computational resources than simpler analysis tools. **Plotly** was used to create interactive visualisations and dashboards. Its strength was providing interactive charts that allowed patterns in ratings, genres, runtime, weight, language and other variables to be explored clearly. Its disadvantage is that creating and customising visualisations requires coding knowledge. Overall, the combination of n8n, Polars, PySpark and Plotly was highly effective because their strengths complemented one another across automation, data preparation, Big Data analysis, machine learning and visualisation. Although each tool has limitations, their combined advantages outweighed the disadvantages and made the selected software appropriate for the TVmaze use case.

Evaluate the Data and Findings Produced and Their Usefulness in Providing Meaningful Insights

Here, we summarized the effective TVmaze data in providing valuable insights into the characteristics of television programs and supporting preferred usage patterns. One of the most important strengths of the extracted data is that it collected variables such as ratings, reach, viewing hours, genre (e.g., comedy, drama), language, country, production network, program status, and year of broadcast. This helped us gain the opportunity to view the data from different angles. Another advantage is that the interactive dashboards made it easier for us to understand the results and compare them. Now, let's take an example.

The analysis showed that most television programs are between 45 and 60 minutes long, and that highly rated programs are in wider demand and reach than those with lower ratings. The language and country analysis also showed a strong influence of English-language content and programs from the United States, providing greater reach. However, we now focus on the main weaknesses of some variables: the presence of unseen or unknown information, resulting in a large unknown category and reducing the reliability of comparisons. Another drawback is that the results we observe mostly describe relationships and patterns rather than demonstrating that a single event can cause a high rating. In addition, some categories were taken on a much larger number of programs compared to other programs, which means that direct comparisons may be affected by differences. Despite this apparent problem, the advantages of the results may outweigh the disadvantages, as the dashboards and analytics provide clear evidence about the characteristics associated with highly rated programs. In general, the data has already been useful enough to support the extraction of valuable insights, and can help organizations make the necessary, more informed decisions about content recommendations and prioritization, provided that multiple variables are considered together rather than relying on a single result.

Evaluate the Overall Impact of the Implementation on Business Decision-Making for the Selected Use Case

We learned that the overall impact is positive for business decision-making when using specific TV program analytics, as TVmaze transforms data into structured information, analytical results, and interactive infographics that support more evidence-based decisions. Among its key strengths are automated data processing via n8n, preparation using Polars, big data analytics and machine learning using PySpark, and interactive graphing via Plotly. This system provides decision-makers with crucial information on characteristics such as ratings, popularity, genre (e.g., comedy, drama), runtime, language, country, production network, television program status, and first air year. Most importantly, decision-makers can use these results to identify characteristics associated with highly rated television programs, giving them greater attention when developing and prioritizing content. Furthermore, the dashboards make it easier to understand more complex results, improving communication between technical and non-technical users. We are talking about the weaknesses of the application, that the results do not guarantee the success of any television program, because the analysis may identify relationships rather than directly proving causes. Another of its drawbacks is that some categories contain undefined information, such as the production network, which may reduce the reliability of some comparisons. In addition, decisions based on the current set of information may change with the emergence of new television programs and higher viewership ratings, which necessitates updating the analysis regularly. Despite the limitations, which are clear to us, the advantages clearly outweigh the disadvantages because the application provides a structured, evidence-based approach that is better than relying solely on predictions or personal judgment. The application can support more informed decisions regarding content recommendations, genre priorities, and, most importantly, which features may deserve more attention, while final business decisions should combine analytical results with other relevant business factors.

Overall Evaluation and Justified Conclusion

Evidence supporting the evaluation comes from implementation itself, the analysis of data collected through API of TVmaze that gave preparation data and created dataset with data variables like rating, runtime, genre, weight, language, country and year of premiere. Data gives immediate evidence for the findings made in the analysis. For instance, the first dashboard shows that most common category of runtime is from 45 minutes till 60 minutes while the weight analysis indicates that the better rated shows can have wider range of popularity weight. The

second dashboard also indicates strong representation in the dataset of English-language shows and shows produced in the USA. Nevertheless, the evidence has some limitations, such as a large Unknown category in terms of production networks and differences in the quantity of records between categories. In addition, the evidence indicates correlation between variables rather than cause-effect relations only. Therefore, the conclusions of the evaluation related to the effectiveness of implementation are supported by evidence obtained from the analysis, and business decisions may need to take more factors into account rather than to base a decision on one finding.

Advantages and Operational Challenges of Using Complex Datasets

For the selected **TV Show Analytics** use case, relying on complex datasets can provide several important advantages for an organisation. First, data-driven decision-making helps decisions match the problems that the organisation has because it relies on real data from the organisation rather than only intuition or assumptions. In this project, analysing weight, genre, runtime, popularity weight, language, and country allows the organisation to identify characteristics that make shows achieve higher-rated shows and use them when considering content recommendations, as they are factors in changing the rating values. Another advantage is gaining a competitive advantage, as analysing trends can help an organisation respond to changes in the TV market more effectively. Complex datasets can also support strategic planning and continuous improvement by identifying trends and areas that require improvement, which makes it easier to understand the pattern

However, using complex datasets also creates several **operational challenges**. Because it needs high effort-time data preparation because large scale data has different values and formats. **Data privacy and security** are also major concerns because the larger the data became the more difficult to apply the security regulations on it to protect it all while it still being updated security breaches can cause financial, legal and reputational damage. Another challenge is **communication with non-technical stakeholders**. If managers cannot understand analytical models or findings, the resulting solutions may not be implemented effectively

Overall, the advantages of complex datasets outweigh the operational challenges when appropriate data preparation which provides more examples and data to understand the patterns under every condition, security controls, and clear communication are applied and implemented to each part of the used data. For the TV show use case, complex data can provide stronger evidence for content recommendations and strategic planning, but the organisation must continuously manage data quality, security and communication to ensure that these benefits are achieved.

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